

INTRODUCTION TO ASTRONOMY



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Introduction to Astronomy

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CHAPTER OVERVIEW

1: Introduction to Astronomy

Learning Objectives

Upon completion of this module, the student will be able to:

- Describe the characteristics of the major classes of objects in the Universe
- Recognize the scale of the Cosmos, using Astronomical Units and Light travel
- Identify specific Constellations
- Describe the characteristics of specific Constellations
- Identify specific Asterisms
- Differentiate between Constellations and Asterisms

This module starts with the question, “What is Astronomy?” Simply stated, Astronomy is the study of the Universe. Astronomers classify objects into various groups based on characteristics and locations in our Universe. Some of these classifications, such as stars, are easy to understand scientifically. Other objects are more difficult to classify, and in some instances, to entirely understand. This module identifies basic objects in our Solar System and those beyond, the overall scale of the Universe, and patterns in the night sky. So welcome to your Universe: a beautiful and incredible place, as you will soon learn.

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1.1: Module Introduction

“I just looked up at a fine twinkling star and thought that a voyager whom I know now many days sail from this coast, might possibly be looking up at that same star with me. ”

–Henry David Thoreau

Personal Log Entry c. 840's

Astronomy is one of the most-fascinating subjects; the Universe in which we live is simply a beautiful and incredible place!

As one of the oldest sciences, astronomy is also one of the fastest-progressing sciences today. In centuries past, people used the stars to tell stories through patterns in the sky. They used the positions of the Sun and Moon to tell time and the seasons, and the positions of the stars to navigate.

With the advent space exploration starting in October 1957, satellites have added to our knowledge of the Universe, from the surface of Mars to the beginning of the Universe itself. It hardly seems a week passes without a new discovery, either from earth-based observations or the myriad of satellites.

Images of the Universe, from galaxies containing billions of stars to the rings of Saturn, inspire both the casual and professional stargazer alike. Astronomy is a passion for many and today is a mainstream hobby, as in years past.

Introduction to Astronomy will provide you with an opportunity to explore both the significant historical aspects of astronomy, as well as contemporary exploration of the Universe. Common topics, such as the phases of the Moon, tides, seasons and the space program, will also be explored.

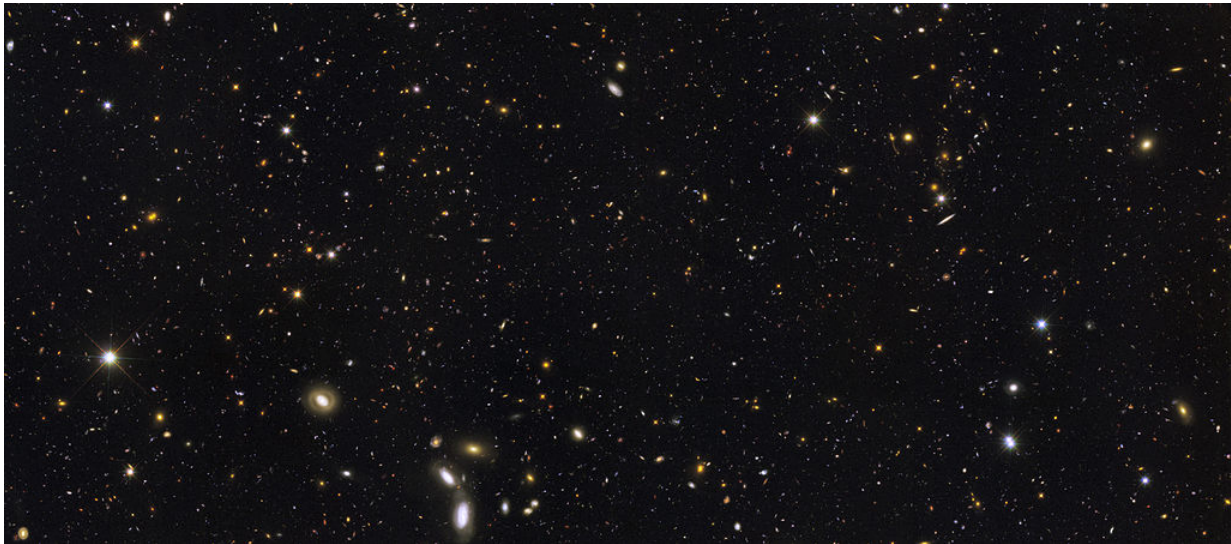
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1.2: Basic Objects in the Universe

This module starts with the question, “What is Astronomy?” Simply stated, Astronomy is the study of the Universe. Astronomers classify objects into various groups based on characteristics and locations in our Universe. Some of these classifications, such as stars, are easy to understand scientifically. Other objects are more difficult to classify, and in some instances, to entirely understand. This module identifies basic objects in our Solar System and those beyond, the overall scale of the Universe, and patterns in the night sky.

Stars

Stars are glowing balls of gas that undergo nuclear fusion; the Sun is a star.

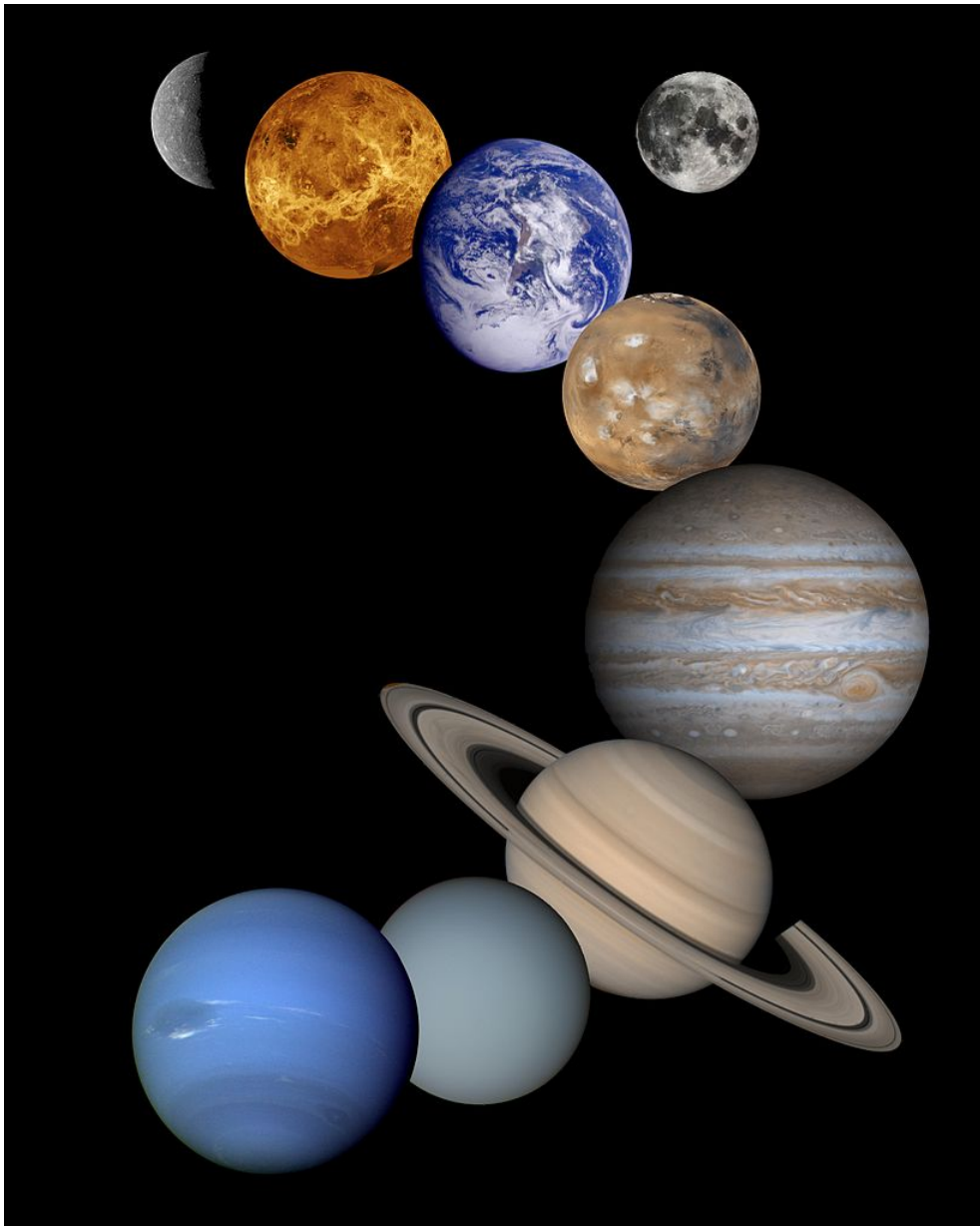


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Planets

Planets are moderately large objects orbiting a star. We see planets because they reflect the light of their central star, or in some cases, stars. Planets are generally rocky or gaseous in nature and spherical-shaped.

A new group of objects has been recently defined: the Dwarf Planets or Plutoids. These are objects that orbit the Sun, but have not cleared their orbits. Pluto is an example of a Dwarf Planet.



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Satellite

A **satellite** orbits a planet; these objects are also called moons. For example, the Earth's satellite is the Moon – a proper name.



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Asteroid

An **asteroid** is a relatively small, rocky/metallic object *usually* orbiting a star.



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Comet

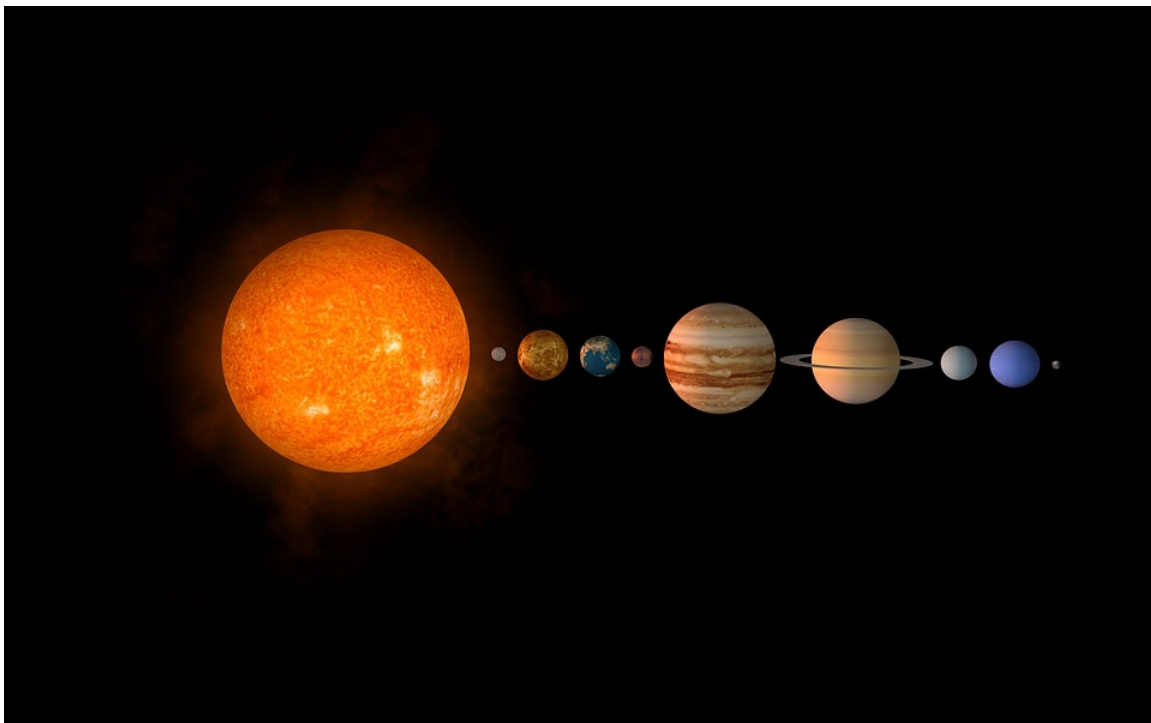
A **comet** is a relatively small, icy object usually orbiting a star. Asteroids, comets, and miscellaneous small/irregular objects and “dust” are often categorized as Minor Bodies.



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Solar System

The **Solar System** is the Sun and all the objects that orbit the Sun, including the planets and their moons.



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Stellar System

A **Stellar System** is a star and other objects such as planets and/or other stars and other materials that orbit it.

Galaxy

A **galaxy** is a large island of stars, a few hundred million to over a trillion stars.



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Galactic Cluster

A **Galactic Cluster** is a collection of galaxies gravitationally bound.

Supercluster

A **Supercluster** is a region where galaxies and galactic clusters are tightly packed.

Universe

The **Universe** is all matter and energy, and is also called the Cosmos.



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1.3: Scale of the Cosmos

Perhaps you have heard the phrase that a large number is **astronomical**. Astronomical comes from the fact that the size of the Universe is so big, it is often hard to comprehend. Even distances between the planets and the Sun in our own Solar System are numbers we are not custom to using. To understand the scale of the Cosmos, we will start with several examples in our Solar System. First, the distance between the Earth and Moon is about 230,000 miles. The closest distance between Earth and Mars is about 32,000,000 miles. The distance between the Earth and Sun is about 93,000,000 miles. The distance between Mercury – the closest planet to the Sun – and the Sun is about 36,000,000 miles. The distance between Neptune – the farthest planet from the Sun – and the Sun is about 2,797,770,000 miles. It is often useful to make a scale model of large systems. Some cities, museums, and parks will create a scale Solar System, based on something like 1 foot equals 1 million miles.

To simplify *distance* in our Solar System, astronomers use the Astronomical Unit (1 AU), which equals 93,000,000 miles or the average distance from Earth to the Sun.

As for other numbers, these are the diameters of three Solar System bodies. First, our Earth's diameter is just less than 8,000 miles. Jupiter's diameter – the largest planet in our Solar System – is around 88,000 miles. And the Sun's diameter is about 850,000 miles.

The idea here is *not* to memorize size and distance; rather, it is to appreciate the grand scale of the Universe within which we live.

Going beyond our Solar System, the closest star is Proxima Centauri, part of the Alpha Centauri triple star system. It is about 24,340,000,000,000 miles (that is 24 trillion, 340 billion miles) away or about 270,000 times more distant than Earth to the Sun. And that is the closest star beyond our Sun. Because of these great distances, astronomers will use another measure: the distance light travels in one year, or the light year. This sounds like a time measure, but it is not.

Light travels 186,000 miles in one second, or 2.99×10^8 meters per second. That is 5,869,713,600,000 miles in a year. So the Sun is about 8 light minutes from Earth and Proxima Centauri about 4.24 light years distant.

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